Base model

Let $p(t)$ and $g(t)$ represent the total arms and military spending in year t by Purple and Green, respectively. The two nations are in competition, so there is an underlying sense of mutual fear: the more one nation arms, the more the other nation is incentivized to arm. To start, we could assume that each nation's armament rate is directly proportional to the other's arms spending (these should be linear terms in your model). We may also want to account for the fact that excessive armament expenditures come at a cost to each nation's economy, i.e., the rate of change of each nation's expenditures is directly and negatively proportional to its own expenditure (again, these should be linear terms in your model). The sign of these constant parameters determines whether the term models grievances, which increase armament spending, or feelings of good will, which decrease it.

Construct a continuous model to describe this scenario. What does your model predict for the long-term arms spending of each country? Notice that analytical discussions will be required to get full score of the this part of the base model for this topic.

Finally, we can additionally model any underlying grievances or feelings of goodwill of each country toward the other using constant terms, which should be invariant to spendings or time. Try to discuss the impact of such goodwills to the model predictions using simulations.

1A

1B

Base Model: Model 1 (do analysis on this model)

★ Model 2 look at this through simulations

Base Model:

$$\begin{cases} \dfrac{dp}{dt} = a\,g(t) - c\,p(t) \\ \dfrac{dg}{dt} = b\,p(t) - d\,g(t) \end{cases} \Rightarrow$$

Linear Autonomous Homogeneous ODE (no constants)

$\Downarrow$

Solvable analytically

- Find equilibrium
- Analyze stability (w/ phase Potrait?)
- Interpet long-term behaviour

$$\frac{d\vec{x}}{dt} = A\vec{x}$$

$$\begin{bmatrix} \dfrac{dp}{dt} \\ \dfrac{dg}{gt} \end{bmatrix} = \begin{bmatrix} a & -c \\ b & -d \end{bmatrix} \begin{bmatrix} p \\ g \end{bmatrix}$$

and Model: (use ★)

$$\begin{cases} \dfrac{dp}{dt} = ag - cp + e \\ \dfrac{dg}{dt} = bp - dg + f \end{cases}$$

Work

- Run simulations